SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	K. Dedeepya	**Reg. No.:**	192324252
Section / Batch:	AIDS SLOt-B	**Date:**	

Code of Conduct

I ___B.Joshna(192472097)_______________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

Instructions

- Read each question carefully before selecting your answer.
- This assessment consists of 50 multiple-choice questions.
- Do not use external resources, AI tools, textbooks, or assistance from others during the assessment unless explicitly permitted by the instructor.
- Upload Score Card in GCR and submit score card printout.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Context-Free Grammars (CFG)	CFG components, productions, terminals, non-terminals, recursion, sentence generation	1 – 10
Sentence Construction and Agreement	Parse trees, sentence structure, grammatical agreement, syntactic validation	11 – 20
Subcategorization	Verb frames, complement requirements, argument structure analysis	21 – 25
Top-Down Parsing	Parsing strategy, derivation process, recursion issues, grammar expansion	26 – 30
Earley Parsing	Predictor, Scanner, Completer operations, ambiguity handling	31 – 35
Feature Structures	Feature representation, unification, constraint checking, agreement enforcement	36 – 40
Probabilistic Context-Free Grammars (PCFG)	Rule probabilities, ambiguity resolution, probabilistic parsing	41 – 48
Integrated Syntax Analysis	CFG + Feature Structures + PCFG integration in NLP applications	49 – 50

Annexure A – MCQ Test

https://forms.gle/WSv28zfaU5B2aWcy6

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Number of Correct Answers	100%	45 to 50 correct answers	38 to 44 correct answers	25 to 37 correct answers	0 to 24 correct answers
Accuracy	100%	Excellent (90–100%)	Good (75–89%)	Average (50–74%)	Poor (Below 50%)

No. of Questions	Number of Correct Answers	Accuracy	Total %
50			

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Y. Varun
192424230

Test

30/50

× 1) Recursive Production rule's allow hierarical sentence → B
× 2) subject verb Agreement → A
× 3) finite state Atomata → C
× 4) Morphological → A
✓ 5) left recursion causing infinite expression → B
✓ 6) [B] user's Predictor, scanner & completion operation.
× 7) Regular expression matching ⇒ A
✓ 8) Choose the Parse tree with most rule Probability [A]
✓ 9) Finite state Transducer [B]
✓ 10) CFG + feature structure + subcategory [A]
× 11) Remove all PP [C]
✓ 12) Authorized employe & update [D]
× 13) [C] The verb Voilates the subcategorization
[E]
✓ 14) [B] left recursion causing infinity recursion
[B]
✓ 15) Dynamic chart Construction Combining scanner & Completer [D]
× 16) Number feature Conflict btwn sub & obj [A]
× 17) Tree 1 because A has ~~initial~~ Probability [A]
✓ 18) Highly recursive grammar with unrestricted Production [A]
✓ 19) [D] Proxy Passing handler
× 20) CFG Parsing - subcategory Checking · [C]

✓21) [C] Introduce a Compound noun Production
✓22) [A] Main subject nounPhase
✓23) [D] verb subcategorie frame
✓24) [A] ~~shift~~ reduce
x25) [A] Prediction
x26) [B] Maximum likely distribution
✓27) [A] Tree A because it has highest probability
✓28) [A] Context free grammer only
✓29) [B] recursive top down Parser
x30) [A] Maximum Frequency matching
✓31) [A] S – NP/VP → on, N, VP – the
x32) [D] Engineer Develop's the software
x33) [C] She gave the Customer
✓34) [D] Expand S into NP NP
✓35) [A] Predictor
✓36) [B] uniform false because feature
✓37) [A] The Phase withe the highest overall
✓38) [B] Recusive decent Parsing
✓39) [A] Phase A because it satisfies the Constants's
✓40) [B] Reject the Phase because grammitical argeement's Contrad
x41) [B] ~~Den~~ NP Det/N
x42) [A] Det
✓43) [C] completer
x44) [A] sentence Accepted because Person matches
x45) [A] The maximum ~~Pro~~ no of Production rules
✓46) [A] CFG supports recursive non Phase generation NP→and; NP
✓47) [C] structred (Symantic) antiguese
x48) [C] Recursive grammer expansion
✓49) [A] The verb gave recursive additional Compenents
✓50) [A] V NP PP